

LATERAL STRENGTH AND DUCTILE BEHAVIOR OF A MORTISE-TENON CONNECTED TIMBER FRAME

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ABSTRACT: The primary goals of this project were to examine the amount of lateral force resisted by a single-bay mortise-tenon connected timber moment frame, and to introduce ductile behavior into the mortise-tenon connections by adding a steel sleeve around a traditional wood peg. This research aimed to provide proof that traditional timber frames are capable of ductile racking while reliably complying with the American Society for Civil Engineers (ASCE 7-10) building code drift specifications, implying an increase in the ASCE 7-10 ductility factor (R) for wood frames when used as lateral force resisting elements. A secondary goal was to promote traditional heavy timber framing as a primary structural system. Modern structural framing is dominated by light-wood, steel, and concrete. The exploration in this project aspires to demonstrate that heavy timber frames can achieve comparable lateral performance and frame behavior to other current lateral systems.

KEYWORDS: Ductile, Mortise, Tenon, Heavy Timber, Seismic Performance, Timber Frame, Dowel

1 INTRODUCTION

This project researched the lateral strength and ductility of a timber post-beam frame with angled kick braces triangulating the top two corners, using traditional heavy timber connections. All joints were connected by inserting a tenon into a mortise pocket and hammering a dowel through pre-drilled holes. The dowel was in double shear as it held the tenon inside the mortise pocket. Heavy timber frames of this configuration have potential to be used as reliable lateral force-resisting elements. As the beam translates laterally, the changes in geometry can be accommodated by isolating damage into the dowels that connect all of the members, primarily the dowels connecting the kick braces to the beam and columns. The forces that generate in the kick braces, as the frame geometry changes while racking, flow to the dowels then into the main member that houses the connection, either the beam or the column. If the dowel holding the connections together is engineered to fail in a safe manner before the main wood members, the frame can accept changes in the geometry (permanent deformations), the dowel can be used as a non-linear plastic hinge element, and in general the frame will dampen structure movement. Control of the

frame's racking behaviour was achieved by designing the dowels to form plastic hinge zones, similar to a reduced-beam-section in steel moment frames, before any of the main timber members fail in these modes: compression parallel and perpendicular to grain, bolt tear out, or tension parallel and perpendicular to grain. As long as the dowel hinging is engaged before the previously listed failure modes, the main structural elements that hold up the building against an earthquake or gust of wind will not fail and collapse.

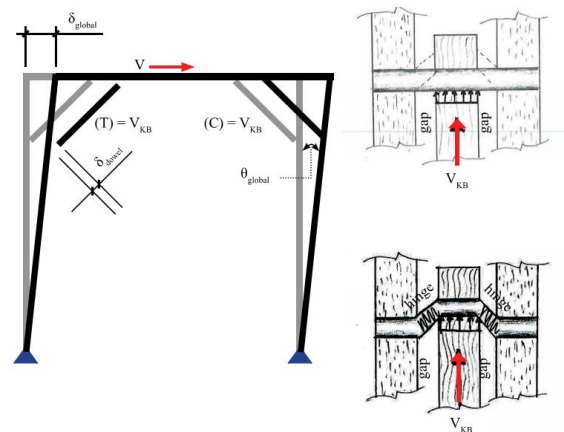


Figure 1: Frame racking and dowel hinge deformation.

2 DOWEL TEST

Heavy timber frames that can resist seismic forces have existed for centuries. Traditionally, wholly wood pegs were used to connect all of the members, but in this research wood pegs were inserted into a metal tube sleeve creating a composite cross section with wood peg insert filling the inside and a thin-walled metal pipe on the outside. The wood peg inserts were used to guide the

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deformed shape of the ductile metal tubes. The metal material is used for slow, inelastic deformation, which translates into energy dissipation, as it controls the global frame action with ductile and stable behaviour. The dowel and the main structural parallel strand lumber (PSL) [14, 15] members were sized such that the engineered timber withstood the forces imposed by the dowels at each connection to the kick braces, forcing the ductile dowel to be the only element that yields. The challenge was engineering the optimum ratio of dowel strength to the strength of the surrounding wood housing (mortise tenon connection), such that wood failure in the main wood members is avoided. As a starting point, a very thin walled metal pipe was used to encase a wood peg. Using thin walled pipes weakened the stiffer metal material by encouraging local wall buckling. The dowel was not expected to fail in shear because local buckling was expected to govern. A gap between the mortise and tenon contact surfaces was detailed to ensure the thin metal pipe walls deform and buckle locally in the desired plastic hinge regions. To achieve this gap a ½" deep countersunk hole on each side of the tenon bolt hole was provided. Including this gap voids the National Design Specifications [3] for Wood Construction provision to analyse different yield modes with dowel connections because the gap forces only one major yield mode, Mode IV [3]. Using a Tinius Olsen Testing Machine, shown the system was loaded in compression at one half of an inch per second. The dowel test and results are given below:



Figure 2: Dowel Testing machine and PSL housing.



Figure 3: Tested stainless steel and wood peg dowel.

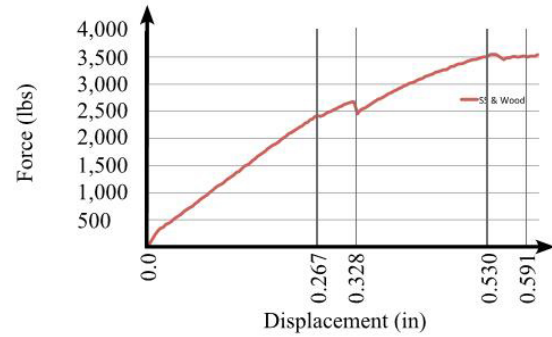


Figure 4: Force versus displacement stainless steel and wood peg dowel.

Table 1: Stainless steel and wood peg compression test values.

Dowel	Yield Load (lbs)	Yield Deflection (in)	Ultimate Load (lbs)	Ultimate Deflection (in)
Stainless Steel	2,500	0.267	3,500	0.591

As the machine compressed some minor cracking could be heard, this was the sound of the test housing settling. Eventually some louder cracks were heard, however, nothing seemed to be happening to the test housing, so it was clear that this noise came from the wood peg insert breaking at the notches. Even though the wood peg inserts broke, the dowel continued to resist increasing loads until around one-half of an inch of deflection. The ultimate load of the dowel was 3,500 pounds, see Table 1. In the dowel hinge locations local wall buckling occurred. The steel sleeve remained smooth at the ends and middle of the dowel. The jumps in the data, see Figure 4, were caused by the wood peg insert fracturing and the metal pipe taking some time to regain stiffness. Once the dowel regained stiffness, the slope of the curve is almost the same as the slope of the data curve before the wood inserts fractured. The steel sleeve did not visibly crack or rupture. Steel has been discovered to retain material strength under many cycles of loading far better than any other metal (Forrest, 1970), making it a preferred building material when it is known the structure will endure fatiguing cyclic loading, also known as wind or earthquake forces. The reason for steel being tough and able to endure cyclic loading is that it is a ferrous metal, meaning it is composed of iron. Metals containing iron have a high fatigue limit, unlike aluminum or copper alloys. This is explained in a paper written solely about fatigue in metals from the Ministry of Technology in England (Forrest, 1970). The composite dowel demonstrated strength and deformation qualities that would result in frame behaviour capacity comparable to plywood shear walls.

3 FRAME THEORY AND ANALYSIS

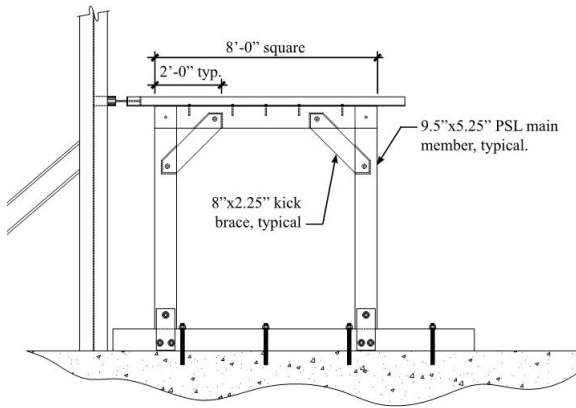


Figure 5: Tested frame configuration.

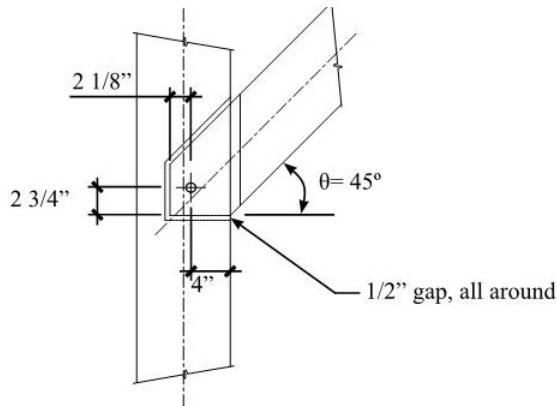


Figure 6: Typical kick brace connection.

3.1 THEORY

The timber frame configuration researched in this project was an eight foot wide by eight foot tall PSL post-beam wood frame with two kick braces triangulating the top two frame corners. True pinned base connections and kick brace connections, and a pinned beam to column connection hold the members in their desired configuration. What makes the frame tested in this project unique compared to traditional timber frames is the use of a steel tube sleeve that encases a wood peg. This engineered dowel, and detailing around that dowel inside of the mortise-tenon connection, allow the composite peg to serve as a “fuse”, isolating permanent frame damage into plastic hinges in order to accommodate changes in frame geometry as the frame racks. The metal pipe is necessary because metal can endure many cycles of fatigue and strain [15], unlike a traditional and brittle wood peg.

3.2 ANALYSIS

Structural analysis was used to predict the capacity of the frame to be tested, both elastic and inelastic frame behavior [8]. Knowing the dowels could safely – ductily – deflect at least one half of an inch, and with a 1/2 inch gap for tolerance in the mortise pocket depth, the total amount the frame could deflect based on geometric translation tolerances was estimated by hand as 1.41

inches. This is a deflection calculation based solely on the geometry of the frame and its connections, without contributions due to members deforming (bending, compressing, stretching) which means the frame will likely deflect more than 1.41 inches. This deflection was matched with a 1,500 pounds ultimate elastic lateral load using RISA 2D 2013. Applying a 180 pounds per foot (plf) load, or 1,400 pounds, laterally to the beam resulted in a force in the kick braces that would yield the dowel based on Table 1. To account for friction in the connections, 1,500 pounds was used as the approximate ultimate force resisted by the frame. Paired with the previously calculated expected lateral deflection, an expected ultimate force versus displacement point called “hand calculated” was plotted on the first test hysteresis, see Figure 10. A nonlinear MATLAB code using Newton Raphson matrix structural analysis was created. Microsoft Excel was used to implement the nonlinear dowel stiffness data into the MATLAB code by solving for an equation that best fit the data curve from the steel sleeve with wood peg insert dowel, see Figure 4. This best fit equation was used in the stiffness function of the code, changing the frame stiffness as the iterations incrementally applied lateral force to the beam. Thirty-Two degrees of freedom were needed to model this frame in MATLAB, see Figure 7. Members one, two, five, six, seven, and nine were modeled as fixed-pinned elements. Member eight was modeled as a fixed-fixed member. Elements four and five were truss elements. Elements ten, eleven, twelve, and thirteen were also truss elements, however, their stiffness was based on the dowel testing data. Figure 8 is a graph of the non-linear pushover curve output by the MATLAB script.

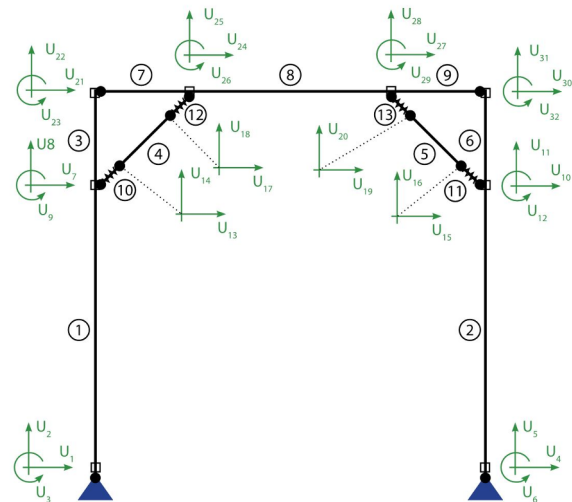


Figure 7: MATLAB degrees of freedom model.

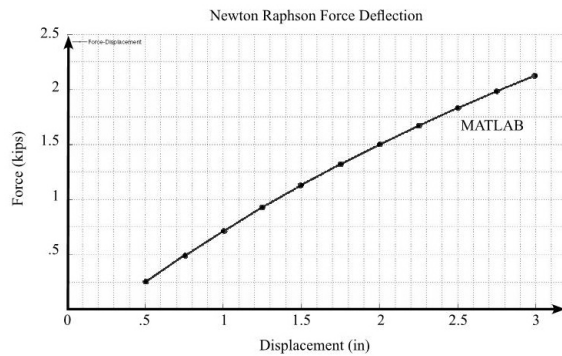


Figure 8: MATLAB predicted non-linear frame behaviour.

The MATLAB code predicted the frame would resist over 2,000 pounds of force at a lateral deflection of three inches.

4 FRAME TEST ONE

CUREE testing protocol for woodframe structures [7] suggest test loading patterns based on load or deformation that simulate earthquake cycles. This testing will not push on the frame dynamically as rapidly as an actual earthquake, but the suggested CUREE test procedure is trusted to produce a quality hysteresis that tests the cyclic and relatively dynamic capabilities of the frame to resist lateral forces.

4.1 TESTING PROCEDURE

The hydraulic press was pumped by hand until the computer read the first data point per the CUREE testing protocol, see Figure 9. One cycle was defined as a full deflection from right to left then back to the starting position. Testing stopped once the largest deflection the ram could extend to (three inches in either direction) was reached, or if the frame failed. The force and displacement data points were recorded every three seconds to extract a hysteresis.

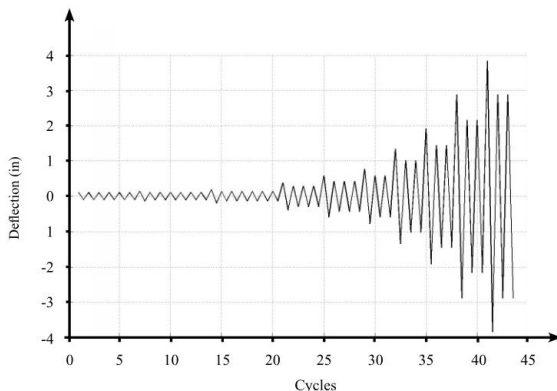


Figure 9: Curee testing displacement graph.

4.2 TESTING OBSERVATIONS

As the frame was racked it rebounded elastically until roughly 600 lbs. As the deflections got larger some rotations occurred at the beam to column connection, and the kick braces either pulled out of or were pushed in to the mortise pocket. Rotations at the

bottom of the columns was also observed. No crushing occurred in the main wood members. Dowel deformation was the most important sign of proper frame behavior. It was observed from the outside that the dowels had been bent and crushed inside of the mortise-tenon connection. The dowels endured sixty cycles of racking.

When the frame racked there were medium-loud cracking noises. It was discovered that these noises came from both the wood peg dowel inserts breaking and friction between the surfaces of the mortise-tenon connections as they rubbed against each other. The connections were so tight tightly fit that it took roughly fifty pounds incrementally to overcome the static friction. When the static friction was eventually overcome, a -medium loud snap or crack sound echoed.

When testing was complete, the base connection was fully intact and showed no signs of weakening. The bottom surface of the column rested parallel to the surface of the sill and rigid foam.

4.3 TEST RESULTS

The maximum load put into the frame was 2,000 pounds with a total lateral deflection of 2.9 inches. The frame endured over sixty cycles of lateral loading. No members showed signs of damage, they could be reused. Δ_u , also shown as Δ_a , is the expected maximum inelastic drift per ASCE 7-10. This is the ultimate allowable inelastic deflection, meaning the lateral system's force-versus-deflection output should be nonlinear at this deflection, calculated to be just under two inches for an eight foot tall frame per Table 12.12-1 in ASCE 7-10. Figure 10 shows that the frame was in the inelastic range at Δ_a .

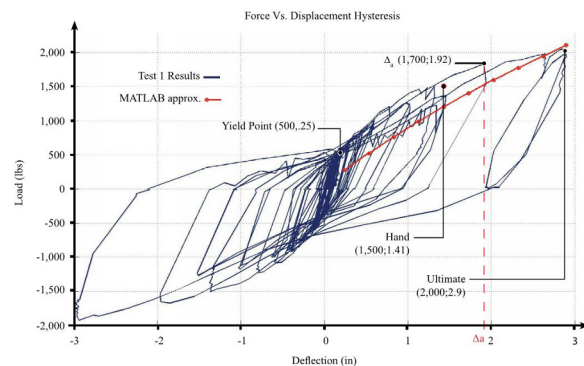


Figure 10: Test one hysteresis.

4.4 HYSTERESIS DISCUSSION

The frame remained elastic until 500 pounds (62.5 plf) with a lateral deflection of 1/2 inch. The inelastic range went up to 2,000 pounds (250 plf) with 3 inches of deflection. The beginning of the hysteresis has a reasonable shape for a frame which has deformation controlled by steel: a straight elastic portion followed by a yield point then a yield plateau of decreasing stiffness shown by a slightly arched shallow angled line. As the load in the frame is reversed, the hysteresis curve is headed to the zero force line at an angle very similar to the initial elastic stiffness slope. However, unlike most steel controlled systems, the force vs. displacement line

loses stiffness, shown by a decrease in slope, as the frame force approaches zero. The change in slope creates a pinch in the overall shape of the graph, similar to the shape of a concrete moment frame hysteresis. The reason concrete moment frame hysteresis graphs have this pinching is because concrete cracks as it bends, changing the moment of inertia of the section. A similar thing is happening to the dowel in the wood frame. As the dowels are pushed in, the hinge locations of the dowel deform the cross-sectional shape, changing the cross-sectional geometry of the region of the dowel that controls lateral movement of the entire frame. Compare this to a reduced beam section in steel moment frame. The cross-sectional geometry of a steel reduced beam section is able to stay fairly constant when it plastically deforms because the web and flanges are designed to be compact, creating a graph that looks like the outline of an American football, at a forty-five degree angle. In this research the plastic hinge region of the dowel deforms into a smaller cross-sectional shape, meaning the hysteresis pinches decreasing energy absorption. As the dowel deforms the controlling hinge section gets pinched, weakening cross-sectional properties. Because of this, the dowel loses its stiffness much sooner than, say, a steel wide flange that can maintain a more constant cross section throughout the nonlinear range.

The benefits of using stainless steel sleeved dowels is demonstrated by the frame enduring 62 cycles, far more than the number of cycles that a timber frame with wholly wood dowels could be expected to endure. A structural engineer seeking to implement a traditional timber frame into a building would not want to be responsible for the liabilities that come with relying on a wholly wood dowel to stabilize a lateral system enduring multiple cycles. Technically, a wood frame with all wood dowels would not have zero capacity after a dowel breaks. As the load reverses there would realistically be some capacity that remains, however the amount of stiffness would be small and unpredictable. The design capacity of a wood dowel that has ruptured should not be relied on and should be taken as zero. The material properties of every wood dowel in a frame also differ from each other because wood is a natural material that humans cannot control. Because of this and because of how little strength wood retains after the fibers have been strained passed their yield or rupture point, only half of an inelastic cycle can reliably dissipate energy. Essentially, when wholly wood dowels are used in a traditional timber frame, the dowels cannot be relied on to endure multiple cycles of lateral loading, so the usable portion of the hysteresis is half of a cycle. When the stainless steel sleeve is introduced into the connection, the benefits of steel's ability to endure multiple cycles are seen in the hysteresis in Figure 10. The frame can sway—also known as rack—back and forth, dissipating energy from not just one but numerous complete cycles imposed by seismic events. Building types that require ductile lateral systems are non-essential facilities, which make up a large amount of buildings. Non-essential facilities are all building other than those that are intended to remain operational in the event of extreme environmental loading from flood, wind, snow, or earthquakes (ASCE, 2010). Non-essential structures are engineered for permanent damage or inelastic

performance design in order to dissipate energy and dampen lateral building movement. The timber frame tested in this project would not be used for essential facilities—buildings that are designed to remain elastic during maximum considered earthquakes. For buildings that do not need to be immediately occupied—non-essential facilities—after a seismic event, multiple cycles of nonlinear frame behavior are desired because energy dissipation is the performance goal.

4.5 TEST ONE SUMMARY

The ultimate total force imposed on to the frame was 2,000 pounds. The maximum lateral deflection was three inches. The frame did satisfy the code requirements of being in the non-linear range at a drift two percent, Δ_a . The support frame deflected 0.036 inches. Based on the support frame deflection measurements all measured test frame deflection values should technically be reduced by 98.8% of the recorded frame deflection, however, this reduction was considered negligible.

5 REUSABILITY

After a successful first test reusability was investigated. This project promotes sustainable building by using wood and that is further reflected within concept of reusability. PSLs were used because of their inherent strengths, but also because of how hard and stiff the material is. The surfaces of most sawn cut wood can be dented with a fingernail, not PSLs. In earlier dowel tests it was seen that the PSL dowel test housing was undamaged around the dowel holes. Just like the dowel test housing, after the first full frame testing had commenced the old damaged dowels were replaced them with new ones, all frame members were reused. The main members were examined and looked adequate to still be used to support a building under gravity loads and even during another earthquake. To confirm this, a second test was conducted. To replace the dowels the frame was brought back to its zero displacement point. In reality, bringing an entire building back to its zero point may have to be done to replace the dowels, but this is feasible and something that is already done today when repairing damaged structures after an event. If the dowels are removed while the frame is in a racked position the entire system could collapse. The frame tested in this project would be able to be re-centered and used again since only the dowels were damaged. Theoretically the primary structural members could be reused multiple times to resist the handful of seismic events or extreme gusts of wind a building has potential to experience in its life time. Upon removal, the dowels were observed to have hinged in the desired locations, and the dowel holes had no visible damage that would significantly reduce the capacity of the frame if another test were conducted. It was concluded that the frame was in an undamaged state and another series of racking cycles was tested.

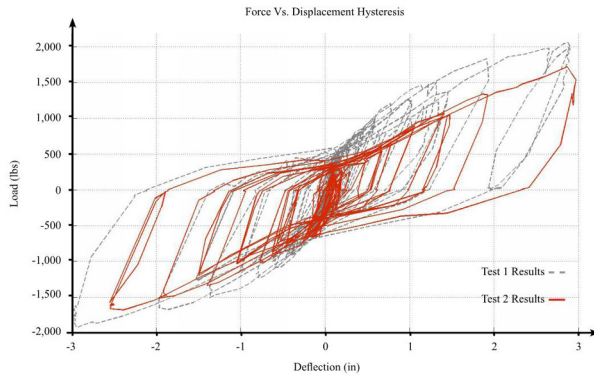


Figure 11: Test two hysteresis.

The second test was almost identical to the first. The frame racked, some snapping noises from movement were heard, as well as cracking of the wood peg dowel inserts. The main members were not damaged.

The test two data is very similar to the test one data set, however it is ten percent weaker. The frame was able to retain ninety percent of its strength reaching a maximum load of 1,800 pounds after the process of replacing the dowels. This reduction in strength came from loosening of the mortise-tenon joints from the first test, and possibly some very slight deformation of the dowel holes from the first test that was not visible to the eye. No further testing was conducted. The results of the Reusability test demonstrate the benefits of having a structural system that can withstand an earthquake by sacrificing a replaceable element. In the case of a heavy timber structure, after an earthquake the main timber members would not have to be replaced or repaired. A team of workers would only have to go around the building and install new dowels, a much easier and cost affective process that installing new structural members.

6 CONCLUSIONS

The lateral strength of the frame tested was 2,000 pounds, or 250 pounds per linear foot (plf) for an eight-foot frame with an aspect ratio of one to one. 250 plf is enough resisted force to prove that timber frames can be used as a lateral force resisting system, however, the frame is not yet highly competitive from purely a strength standpoint compared to other current lateral systems. If the frame were implemented into building codes a factor of safety would likely be required, reducing the maximum expected capacity of the frame, however, 2,000 pounds of resisted lateral load is enough to demonstrate that timber frames with mortise-tenon-dowel connections are very capable of being reliable lateral systems for residential and small commercial buildings in seismic categories A through C. The frame also retained ninety percent of its strength during the second test exemplifying the frame's ability to retain stiffness after almost 200 cycles, making reusability of heavy timber framing a possibility.

Ductile frame behavior was evident during testing, and is displayed in the hysteresis graphs, Figure 10 and Figure 11. Ductility in a lateral resisting element is directly related to the amount of energy dissipated. The amount of energy dissipated is calculated by quantifying the area inside of a hysteresis loop, per

frame cycle. Between the number of trial and recorded test cycles, the frame was racked back and forth over sixty times during both the first and second full scale tests. No PSL members were considerably damaged after two tests. The only significant failures occurred in the isolated stainless steel sleeved dowels, condoning energy-dissipating frame racking. Two last features that can make the frame tested in this thesis desirable to structural engineers are a hinging sequence and an increase in the R value for timber frames (ASCE 7-10). Not all the dowels engaged at once during both tests. Not engaging all hinges at the same time gives opportunity to create a hinging sequence, creating redundancy in the lateral element. With 6 dowels per frame, a complete structure composed of the system tested in this research would be filled with redundancies. With properly designed mortise-tenon connections the hinges could be timed to yield in a particular sequence. In ASCE 7-10, Table 12.2-1 the R value given for a wood frame when used as the lateral system is 1.5. Ductile wood frame behavior shown in this research suggests an increase in the R value for timber frames when ductile dowels are introduced to the connections. Design base shears for a building using the wood frame tested in this research should not be penalized by the low ductility factor of 1.5. Knowing the structural system is very ductile and redundant gives designers more confidence in the reliability of the building's performance during a seismic event. Normally, structures that rely on wood as a source of ductility are considered to be very brittle, so they are penalized by a low R value, which increases the design base shear, increasing lateral demands, which translates to economical and architectural costs. Increasing the R value for timber frames would help the wood industry; designers would consider using a wood structural system more often. Applications of this research could be used in new construction, or retrofitting soft story buildings that have open garage spaces on the first floor and living spaces above.

A post-beam timber frame with kick braces attached by a mortise-tenon-dowel connection is capable of complying with current minimum design loads per the American Society for Civil Engineers (ASCE 7-10). With modern analysis and design theories, timber frames can be improved to be used as reliable lateral force resisting structural elements, as well as satisfy architectural demands. The advantages of using the metal tube dowels are that they can act as a "fuse" and not only can be replaced, but also allow all other members to remain elastic and reusable after an event. Not only that, but because of metal material properties, dowel deformation inhibits ductile frame action which is important for owners and occupants of a building. Both the MATLAB code created in this project and the structural analysis of wood frames paper from Stanford [8] demonstrates the ability to accurately analyse a wood frame's behavior and ability to resist seismic demands. The benefits discussed regarding wood being cheaper, more sustainable, and aesthetically warm and pleasing emphasize the need to construct buildings with wood. Wood working and hands on construction/fabrication can be an art form. Some of the oldest and most inspiring structures in the world use timber as a building

material. With new energy demands, the building industry is being pushed to shift back towards wood construction and adapt timber structural designs to modern structural and architectural demands. Further testing can explore stiffer dowels, pushing the boundaries of wood and improving energy dissipation. More parameters to test are: a thicker steel sleeve, different dowel hinge designs (see Figures 12, 13, and 14), different frame dimensions and aspect ratios, tighter-fit mortise tenon connections with no rigid insulation, a smaller or no countersink on the kick braces, adding kick braces to the bottom corners, and alternate base connections. Each of these parameters should be tested individually, altered one at a time to clearly demonstrate their effects on the overall frame behaviour. Engineering a hinging sequence can provide a performance-based design with this timber frame. Further testing shall aim to make a traditional timber frame with steel sleeved dowels highly competitive amongst modern framing systems.

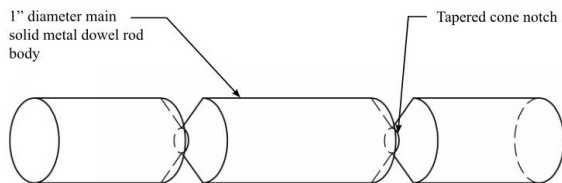


Figure 12: Cone hinge dowel.

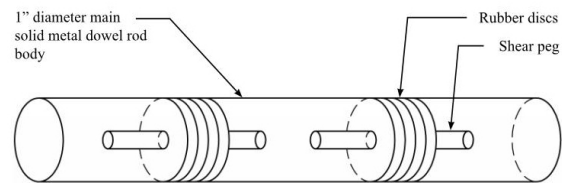


Figure 13: Disk hinge dowel.

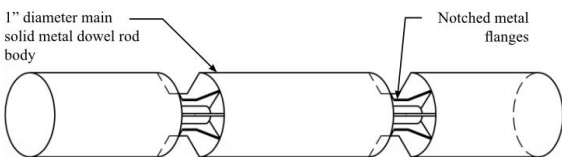


Figure 14: Flange hinge dowel.

ACKNOWLEDGMENT

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Garret McElveny is a Project Engineer at Taylor & Syfan in San Luis Obispo, CA, and a former employee at Dos Osos Timber Works Incorporated. Mr. McElveny is passionate about timber framing and believes that timber frames can be more commonly used as reliable framing systems in seismic zones. Mr. McElveny introduced the initial subject and contributed his experience with timber structure to guide the growth of concepts in this research. Hayward Lumber is a timber distribution company in San Luis Obispo, CA. Although they did not offer discounts for educational research projects, their services allowed completion of this project within a reasonable budget.

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